COMPREHENSIVE FIELD MANUAL

TKanAI SDLC Ecosystem

Complete Role-Based Workflow Guide & Standard Operating Procedures

PLATFORM	ARCHITECTURE	SIMULATION SCENARIO	SUPPORTED ROLES
TKanAI Beta v1.0	**Client-Side IndexedDB**	**Zyroute Bus System**	**10 Personas**

1. System Architecture & Getting Started

TKanAI is an offline-capable, client-side agile management and AI engineering ecosystem. It requires zero cloud databases or API subscriptions—everything executes securely inside your web browser via Dexie.js (IndexedDB) with cross-tab real-time reactivity via the `BroadcastChannel` API.

How to Launch the System

1. Open a terminal in the project directory: `cd "c:\Users\ADMIN\Desktop\AI Ecosystem"`
2. Run the local development server: `npm run dev`
3. Open your browser to: **`http://localhost:5173/`**

> **Database Cache & Name Updates**
> If you modified team names in `src/db/seedData.ts`, your browser will still hold previous IndexedDB data. Go to **System Settings** in the sidebar and click **Reset Data** to refresh all records instantly.

Instant Role Switching

To simulate any role, click your **profile avatar** in the top-right header and select from the 10 personas. The UI views, action authorizations, and notifications adapt immediately without a logout.

2. End-to-End SDLC Project Workflow

Every project in TKanAI follows a structured 8-stage lifecycle. Each stage is driven by specific personas using specialized AI copilots.

1 Project Initiation & Scope

Roles: Org Admin (Vanuya), PM (Thivvikan)
Create project portfolio, define sprint cadences, and assemble cross-functional rosters.

2 Requirements & Ambiguity Intelligence

Roles: BA (Kanushi), PO (Ushamini), Client (Sutharsini)
Draft FR/NFRs, execute AI Ambiguity Scoring, and conduct formal approvals.

3 Backlog Breakdown & User Stories

Roles: PO (Ushamini), BA (Kanushi)
Use the AI Story Generator to convert requirements into user stories with Gherkin acceptance criteria.

4 Sprint Planning & Kanban Execution

Roles: PM (Thivvikan), Dev (Kajakaran), Designer (Sarangn)
Commit stories to sprints, auto-decompose into subtasks, and advance cards through Kanban swimlanes.

5 QA Testing & AI Bug Lab

Roles: QA Lead (Mayoory), Dev (Kajakaran)
Author test suites, record execution results, and trigger AI Root Cause Analysis (RCA) on defects.

6 Meetings & AI Decision Extractor

Roles: All Team Members
Log sprint ceremonies and use AI to extract Architectural Decision Records (ADRs) and action items.

7 Client Portal & Change Requests (CR)

Roles: Client (Sutharsini), PM (Thivvikan)
Submit scope change requests and run AI Impact Analysis estimating schedule, cost, and test risks.

8 Risk Radar & Executive Delivery

Roles: PM (Thivvikan), Stakeholder (Hariarthan)
Monitor the AI Risk Radar probability matrix and generate printable executive reports.

> **Living Traceability Guarantee**
>
> Every artifact connects end-to-end: `Requirement` ➔ `User Story` ➔ `Task` ➔ `QA Test Case` ➔ `Defect`. Open the **Traceability Matrix** view at any time to verify 100% test coverage and ensure zero orphan requirements.

3. Role-by-Role Step-by-Step Practical Guides

1. Organization Administrator

Persona: Vanuya

Primary Views: Org Dashboard, Project Portfolio, System Settings, Global Search

Mission: Oversee enterprise health, instantiate projects, configure governance parameters, and maintain database integrity.

1. **Step 1 — Switch Role:** Click the top-right avatar and switch to **Vanuya (Org Admin)**.
2. **Step 2 — Inspect Org Health:** Go to **Organization Dashboard** to view macro KPIs: total projects, completed tasks, open critical bugs, and overall document quality score.
3. **Step 3 — Create a New Project:** Open **Project Portfolio**, click **"Create Project"**, specify project key, title, deadline, and assigned leads.
4. **Step 4 — Database Backup:** Navigate to **System Settings** and click **"Export Database"** to save a complete JSON backup of the organization state.

2. Project Manager (PM)

Persona: Thivvikan

Primary Views: Kanban Board, Sprint Analytics, AI Risk Radar, Meetings, Change Requests

Mission: Manage sprint cadences, balance team workloads, evaluate client change requests, and mitigate project delivery risks.

1. **Step 1 — Monitor Active Sprint:** Open **Sprint Analytics** to review the burndown velocity chart, remaining story points, and sprint deadline.
2. **Step 2 — Orchestrate the Kanban Board:** Go to **Kanban Board**. Drag cards between swimlanes (`To Do` ➔ `In Progress` ➔ `Code Review` ➔ `QA Ready` ➔ `Done`). PMs have unrestricted rights to reassign or move any card.
3. **Step 3 — Evaluate Client Change Requests:** When the Client submits a Change Request (CR-001), open the CR detail view, review the **AI Impact Analysis** (cost, delay, affected tasks), and formally approve or reject with decision notes.
4. **Step 4 — Review the AI Risk Radar:** Open **AI Risk Radar** to review high-impact risks (e.g., concurrency race conditions, developer overallocation) and assign mitigation tasks.

3. Product Owner (PO)

Persona: Ushamini

Primary Views: Requirements & AI, Backlog & Stories, Documentation Hub

Mission: Define product vision, prioritize sprint backlogs, approve requirement specifications, and validate deliverable scope.

1. **Step 1 — Review & Approve Requirements:** Open **Requirements & AI**. Review requirements drafted by the BA. Click **"Approve Requirement"** to lock the scope for engineering.
2. **Step 2 — Prioritize the Product Backlog:** Open **Backlog & Stories**. Drag and reorder user stories to define sprint priorities. Ensure all high-priority stories have estimated story points.
3. **Step 3 — Generate User Stories via AI:** In the Requirements view, select a requirement and click **"AI Story Generator"** to convert technical clauses into standard user stories with acceptance criteria.
4. **Step 4 — Approve Living Specifications:** Navigate to **Documentation Hub**, review SRS v1.2, and provide formal Product Owner sign-off.

4. Business Analyst (BA)

Persona: Kanushi

Primary Views: Requirements & AI, Traceability Matrix, Documentation Hub

Mission: Gather functional/non-functional requirements, eliminate specification ambiguity, and link user stories to business goals.

1. **Step 1 — Author Requirements:** Open **Requirements & AI** and click **"Create Requirement"**. Specify category (FR/NFR/Security), priority, description, and acceptance criteria.
2. **Step 2 — Run AI Ambiguity Analyzer:** Select any requirement and click **"Analyze Ambiguity"**. The AI inspects for vague terms (e.g., "fast", "user-friendly") and provides actionable, measurable rewrites with a confidence score.
3. **Step 3 — Inspect Traceability:** Open **Traceability Matrix** to check that every requirement is linked downstream to at least one User Story and QA Test Case.
4. **Step 4 — Maintain Documentation:** Use **Documentation Hub** to update the Software Requirements Specification (SRS) in markdown format.

5. Senior Developer

Persona: Kajakaran

Primary Views: Kanban Board, Issues & AI Bug Lab, Documentation Hub

Mission: Implement features, resolve defects, conduct code reviews, and author architecture decision records.

1. **Step 1 — Check Assigned Tasks:** Open **Kanban Board** and locate tasks assigned to you. Drag your active task from `To Do` to `In Progress`.
2. **Step 2 — Decompose Stories into Tasks:** In **Backlog & Stories**, use the **AI Task Decomposer** to split user stories into backend, frontend, and database subtasks.
3. **Step 3 — Resolve Bugs with AI Root Cause Analysis:** Open **Issues & AI Bug Lab**. Select an assigned bug (e.g., BUG-001). Click **"AI Root Cause Analysis"** to view code fix recommendations (e.g., Redis distributed lock with Lua script). Implement the fix and mark the bug as **Resolved**.
4. **Step 4 — Advance to Code Review:** Move your completed task card to the `Code Review` column on the Kanban board.

6. QA Engineer

Persona: Mayoory

Primary Views: QA Test Suites, Issues & AI Bug Lab, Traceability Matrix

Mission: Author test cases, execute manual/automated regression runs, report bugs, and perform final defect closure.

1. **Step 1 — Execute Test Suites:** Open **QA Test Suites**. Select a suite (e.g., Concurrency & Checkout Suite). Execute test cases and record results: **Passed**, **Failed**, or **Blocked**.
2. **Step 2 — Auto-File Bug on Test Failure:** When a test case fails, click **"Log Bug from Failure"**. The system automatically populates a defect ticket linked directly to the failed test case.
3. **Step 3 — Verify & Close Resolved Bugs:** In **Issues & AI Bug Lab**, filter for bugs in `Resolved` status. Re-test the fix. If successful, perform the QA sign-off to mark the bug as **Closed**.
4. **Step 4 — Kanban QA Sign-off:** Move verified feature cards from `QA Ready` to `Done` on the Kanban board.

7. UI/UX Designer

Persona: Sarangn

Primary Views: Documentation Hub, Kanban Board, Issues & AI Bug Lab

Mission: Design accessible interfaces, maintain design tokens, and review UI implementations against wireframes.

1. **Step 1 — Document Design Guidelines:** In **Documentation Hub**, author and update Design System Specs, interactive wireframe mockups, and mobile responsive guidelines.
2. **Step 2 — Track Design Tasks:** Move design tasks (e.g., "Interactive 2D Bus Seat Selector") on the **Kanban Board** to keep engineering aligned.
3. **Step 3 — File UI/UX & Accessibility Defects:** Open **Issues & AI Bug Lab** to report visual layout bugs, contrast issues, or touch-target defects.

8. DevOps Engineer

Persona: Karikaalan

Primary Views: Kanban Board, AI Risk Radar, Documentation Hub

Mission: Ensure pipeline reliability, infrastructure scalability, database indexing, and automated deployment stability.

1. **Step 1 — Infrastructure Execution:** On the **Kanban Board**, track and complete infrastructure tasks (e.g., Redis cluster configuration, database query indexing).
2. **Step 2 — Maintain Runbooks:** Author deployment runbooks and rollback procedures in the **Documentation Hub**.
3. **Step 3 — Support Risk Mitigations:** Review technical items on the **AI Risk Radar** to assist developers in resolving server-side bottlenecks.

9. Client / Sponsor

Persona: Sutharsini

Primary Views: Client Portal, Requirements & AI, Executive Reports

Mission: Monitor milestone progress, approve project deliverables, and submit change requests (CR) with transparent impact modeling.

1. **Step 1 — Access the Client Portal:** Click **Client Portal** in the sidebar to review overall project delivery progress and active milestones.
2. **Step 2 — Submit a Change Request:** Click **"Submit Change Request"** (e.g., Requesting a 2-Hour Ticket Cancellation Rule). State the business justification and desired deadline.
3. **Step 3 — Review AI Impact Analysis:** View the automatically generated impact breakdown: affected requirements, development hours, and estimated cost adjustments.
4. **Step 4 — Milestone Sign-off:** Formally accept delivery milestones once demonstrated during sprint reviews.

10. Executive Stakeholder

Persona: Hariarthan

Primary Views: Org Dashboard, Sprint Analytics, AI Risk Radar, Executive Reports

Mission: High-level organizational governance, sprint velocity oversight, financial ROI tracking, and executive sign-off.

1. **Step 1 — Review High-Level Dashboards:** Open **Organization Dashboard** for enterprise-level delivery metrics and team workloads.
2. **Step 2 — Track Velocity Trends:** Open **Sprint Analytics** to inspect velocity across historical sprints.
3. **Step 3 — Inspect Critical Risks:** Open **AI Risk Radar** to ensure high-severity risks have active mitigation strategies.
4. **Step 4 — Generate Executive Reports:** Navigate to **Executive Reports** to preview and print clean PDF summaries.

4. TKanAI Copilot & Power Features

Interactive AI Assistant Drawer

Click the ✨ **AI Assistant** icon in the top header on any page to open the **TKanAI Project Copilot**. The copilot possesses real-time awareness of all IndexedDB entities.

User Prompt / Query	Copilot Action & Intelligence
"*What is the current project status?*"	Summarizes active sprint burndown, total story points, open defects, and overall schedule compliance.
"*Which tasks are delayed or blocked?*"	Scans Kanban tasks, filters for blocker flags or approaching deadlines, and alerts assignees.
"*What is our highest risk item?*"	Retrieves the highest-scoring risk from the Risk Radar and recites the active mitigation plan.
"*How is developer workload distributed?*"	Analyzes active task count and remaining hours per engineer to prevent burnout.

Universal Spotlight Search (Ctrl+K)

Press `Ctrl + K` (or `Cmd + K` on macOS) to invoke universal spotlight search. Jump instantly to any task (`TASK-101`), requirement (`REQ-001`), bug (`BUG-001`), or document.

5. Comprehensive Role Permissions Matrix

Permission Capability	Admin	PM	PO	BA	Dev	QA	Design	DevOps	Client	Stakeholder
Manage Org & Users	**YES**	NO	NO	NO	NO	NO	NO	NO	NO	NO
Create Project	**YES**	**YES**	NO	NO	NO	NO	NO	NO	NO	NO
Approve Requirements	**YES**	**YES**	**YES**	NO	NO	NO	NO	NO	**YES**	NO
Manage Backlog / Sprints	**YES**	**YES**	**YES**	**YES**	NO	NO	NO	NO	NO	NO
Move Any Kanban Task	**YES**	**YES**	NO	NO	Own	Own	Own	Own	NO	NO
Execute Test Suites	**YES**	**YES**	NO	NO	NO	**YES**	NO	NO	NO	NO
Verify / Close Bugs	**YES**	**YES**	NO	NO	NO	**YES**	NO	NO	NO	NO
Submit Change Requests	**YES**	NO	NO	NO	NO	NO	NO	NO	**YES**	NO
Evaluate & Decide CRs	**YES**	**YES**	**YES**	NO	NO	NO	NO	NO	NO	NO
Access Client Portal	**YES**	**YES**	NO	NO	NO	NO	NO	NO	**YES**	NO

6. Quick Troubleshooting & FAQ

Q: Why does my browser still show old team member names?
A: IndexedDB caches data after the initial load. Go to **System Settings** in the sidebar, click **Reset Data**, and confirm. All updated names will appear immediately.

Q: Can multiple team members test this together?
A: Yes! Open two browser tabs side-by-side (e.g., Tab 1 as *Developer Kajakaran* and Tab 2 as *QA Mayoory*). When Kajakaran moves a card or resolves a bug in Tab 1, Mayoory's tab in Tab 2 updates in real time via BroadcastChannel.

Q: How do I export or backup my work?
A: In **System Settings**, click **Export Database** to save a complete JSON archive of all projects, requirements, tasks, bugs, and meetings.

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